

ASSIGNMENT - 8.5

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Task Description #1 (Username Validator – Apply AI in Authentication Context)

- Task: Use AI to generate at least 3 assert test cases for a function `is_valid_username(username)` and then implement the function using Test-Driven Development principles.
- Requirements:
 - o Username length must be between 5 and 15 characters.
 - o Must contain only alphabets and digits.
 - o Must not start with a digit.
 - o No spaces allowed.

Example Assert Test Cases:

```
assert is_valid_username("User123") == True
```

```
assert is_valid_username("12User") == False
```

```
assert is_valid_username("Us er") == False
```

Expected Output #1:

- Username validation logic successfully passing all AI-generated test cases.

CODE:

```
workspace > username_validator.py > is_valid_username
1  """
2  Username Validator - Test-Driven Development Implementation
3  Task: Validate username based on specific requirements
4  Requirements:
5      - Length: 5-15 characters
6      - Characters: Only alphabets and digits
7      - Must not start with a digit
8      - No spaces allowed
9  """
10
11
12 def is_valid_username(username):
13     """
14     Validates if a username meets all requirements.
15
16     Args:
17         username (str): The username to validate
18
19     Returns:
20         bool: True if valid, False otherwise
21     """
22     # Check if username is a string
23     if not isinstance(username, str):
24         return False
25
26     # Check length requirement (5-15 characters)
27     if len(username) < 5 or len(username) > 15:
28         return False
29
30     # Check if it starts with a digit
31     if username[0].isdigit():
32         return False
33
34     # Check if it contains only alphabets and digits (no spaces or special characters)
35     if not username.isalnum():
36         return False
37
38     return True
```

```

# =====
# TEST CASES - Generated using TDD principles
# =====

print("=" * 70)
print("RUNNING USERNAME VALIDATOR TEST CASES")
print("=" * 70)

# TEST CASE 1: Valid usernames
print("\n[TEST SET 1] Valid Usernames")
assert is_valid_username("User123") == True, "Valid: 'User123' should be True"
print("✓ assert is_valid_username('User123') == True")

assert is_valid_username("Admin") == True, "Valid: 'Admin' should be True"
print("✓ assert is_valid_username('Admin') == True")

assert is_valid_username("John2024") == True, "Valid: 'John2024' should be True"
print("✓ assert is_valid_username('John2024') == True")

assert is_valid_username("alice42") == True, "Valid: 'alice42' should be True"
print("✓ assert is_valid_username('alice42') == True")

assert is_valid_username("Test5") == True, "Valid: 'Test5' (min length) should be True"
print("✓ assert is_valid_username('Test5') == True")

assert is_valid_username("ValidUser12345") == True, "Valid: 'ValidUser12345' (max length) should be True"
print("✓ assert is_valid_username('ValidUser12345') == True")

# TEST CASE 2: Invalid - Starts with digit
print("\n[TEST SET 2] Invalid - Starts with Digit")
assert is_valid_username("12User") == False, "Invalid: '12User' starts with digit"
print("✓ assert is_valid_username('12User') == False")

assert is_valid_username("1Admin") == False, "Invalid: '1Admin' starts with digit"
print("✓ assert is_valid_username('1Admin') == False")

assert is_valid_username("9john") == False, "Invalid: '9john' starts with digit"
print("✓ assert is_valid_username('9john') == False")

```

```

88 print("✓ assert is_valid_username('User Name') == False")
89
90 assert is_valid_username("Admin 123") == False, "Invalid: 'Admin 123' contains space"
91 print("✓ assert is_valid_username('Admin 123') == False")
92
93
94 # TEST CASE 4: Invalid - Too short (less than 5 characters)
95 print("\n[TEST SET 4] Invalid - Too Short")
96 assert is_valid_username("User") == False, "Invalid: 'User' is too short (4 chars)"
97 print("✓ assert is_valid_username('User') == False")
98
99 assert is_valid_username("a") == False, "Invalid: 'a' is too short (1 char)"
100 print("✓ assert is_valid_username('a') == False")
101
102 assert is_valid_username("ab12") == False, "Invalid: 'ab12' is too short (4 chars)"
103 print("✓ assert is_valid_username('ab12') == False")
104
105
106 # TEST CASE 5: Invalid - Too long (more than 15 characters)
107 print("\n[TEST SET 5] Invalid - Too Long")
108 assert is_valid_username("ValidUsername123456") == False, "Invalid: too long (19 chars)"
109 print("✓ assert is_valid_username('ValidUsername123456') == False")
110
111 assert is_valid_username("ThisIsAVeryLongUsername") == False, "Invalid: too long (23 chars)"
112 print("✓ assert is_valid_username('ThisIsAVeryLongUsername') == False")
113
114
115 # TEST CASE 6: Invalid - Special characters and symbols
116 print("\n[TEST SET 6] Invalid - Special Characters")
117 assert is_valid_username("User@123") == False, "Invalid: 'User@123' contains @"
118 print("✓ assert is_valid_username('User@123') == False")
119
120 assert is_valid_username("User-Name") == False, "Invalid: 'User-Name' contains hyphen"
121 print("✓ assert is_valid_username('User-Name') == False")
122
123 assert is_valid_username("User.123") == False, "Invalid: 'User.123' contains dot"
124 print("✓ assert is_valid_username('User.123') == False")
125
126 assert is_valid_username("User_Name") == False, "Invalid: 'User_Name' contains underscore"
127 print("✓ assert is_valid_username('User_Name') == False")
128

```

```

129 assert is_valid_username("User#123") == False, "Invalid: 'User#123' contains #"
130 print("✓ assert is_valid_username('User#123') == False")
131
132
133 # TEST CASE 7: Invalid - Non-alphabetic starting characters
134 print("\n[TEST SET 7] Invalid - Non-alphabetic Start")
135 assert is_valid_username("_User123") == False, "Invalid: '_User123' starts with underscore"
136 print("✓ assert is_valid_username('_User123') == False")
137
138 assert is_valid_username("-Admin") == False, "Invalid: '-Admin' starts with hyphen"
139 print("✓ assert is_valid_username('-Admin') == False")
140
141
142 # TEST CASE 8: Edge cases
143 print("\n[TEST SET 8] Edge Cases")
144 assert is_valid_username("") == False, "Invalid: empty string"
145 print("✓ assert is_valid_username('') == False")
146
147 assert is_valid_username("ALLCAPS") == True, "Valid: 'ALLCAPS' (all uppercase)"
148 print("✓ assert is_valid_username('ALLCAPS') == True")
149
150 assert is_valid_username("allsmall") == True, "Valid: 'allsmall' (all lowercase)"
151 print("✓ assert is_valid_username('allsmall') == True")
152
153 assert is_valid_username("MixedCase123") == True, "Valid: 'MixedCase123' (mixed case)"
154 print("✓ assert is_valid_username('MixedCase123') == True")
155
156
157 # =====
158 # TEST SUMMARY
159 # =====
160
161 print("\n" + "=" * 70)
162 print("ALL TESTS PASSED SUCCESSFULLY! ✓")
163 print("=" * 70)
164 print("\nValidation Rules Verified:")
165 print("  ✓ Length validation (5-15 characters)")
166 print("  ✓ Character validation (only alphabets and digits)")
167 print("  ✓ Starting character validation (must not start with digit)")
168 print("  ✓ No spaces allowed")
169 print("  ✓ No special characters allowed")
170 print("\n")

```

OUTPUT:

```
=====
[TEST SET 3] Invalid - Contains Spaces
✓ assert is_valid_username('Us er') == False
✓ assert is_valid_username('User Name') == False
✓ assert is_valid_username('Admin 123') == False

[TEST SET 4] Invalid - Too Short
✓ assert is_valid_username('User') == False
✓ assert is_valid_username('a') == False
✓ assert is_valid_username('ab12') == False

[TEST SET 5] Invalid - Too Long
✓ assert is_valid_username('ValidUsername123456') == False
✓ assert is_valid_username('ThisIsAVeryLongUsername') == False

[TEST SET 6] Invalid - Special Characters
✓ assert is_valid_username('User@123') == False
✓ assert is_valid_username('User-Name') == False
✓ assert is_valid_username('User.123') == False
✓ assert is_valid_username('User_Name') == False
✓ assert is_valid_username('User#123') == False

[TEST SET 7] Invalid - Non-alphabetic Start
✓ assert is_valid_username('_User123') == False
✓ assert is_valid_username('-Admin') == False

[TEST SET 8] Edge Cases
✓ assert is_valid_username('') == False
✓ assert is_valid_username('ALLCAPS') == True
✓ assert is_valid_username('allsmall') == True
✓ assert is_valid_username('MixedCase123') == True

=====
ALL TESTS PASSED SUCCESSFULLY! ✓
=====

Validation Rules Verified:
✓ Length validation (5-15 characters)
✓ Character validation (only alphabets and digits)
✓ Starting character validation (must not start with digit)
✓ No spaces allowed
✓ No special characters allowed
```

Task Description #2 (Even–Odd & Type Classification – Apply AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function `classify_value(x)` and implement it using conditional logic and loops.
- Requirements:

- o If input is an integer, classify as "Even" or "Odd".
- o If input is 0, return "Zero".
- o If input is non-numeric, return "Invalid Input".

Example Assert Test Cases:

```
assert classify_value(8) == "Even"
```

```
assert classify_value(7) == "Odd"
```

```
assert classify_value("abc") == "Invalid Input"
```

Expected Output #2:

- Function correctly classifying values and passing all test cases.

CODE:

```
1 """
2 Even-Odd & Type Classification - Test-Driven Development Implementation
3 Task: Classify values based on type and mathematical properties
4 Requirements:
5 - If input is an integer, classify as "Even" or "Odd"
6 - If input is 0, return "Zero"
7 - If input is non-numeric, return "Invalid Input"
8 """
9
10
11 def classify_value(x):
12     """
13     Classifies a value based on its type and mathematical properties.
14
15     Args:
16         x: The value to classify (can be any type)
17
18     Returns:
19         str: Classification result ("Even", "Odd", "Zero", or "Invalid Input")
20     """
21     # Check if input is a boolean (must be done before checking int,
22     # since bool is a subclass of int in Python)
23     if isinstance(x, bool):
24         return "Invalid Input"
25
26     # Check if input is an integer
27     if isinstance(x, int):
28         # Handle the special case of zero
29         if x == 0:
30             return "Zero"
31         # Classify as Even or Odd
32         elif x % 2 == 0:
33             return "Even"
34         else:
35             return "Odd"
36
37     # If input is not an integer, return Invalid Input
38     return "Invalid Input"
39
40
```

```

42 # TEST CASES - Generated using TDD principles
43 # =====
44
45 print("-" * 70)
46 print("RUNNING EVEN-ODD & TYPE CLASSIFICATION TEST CASES")
47 print("-" * 70)
48
49 # TEST CASE 1: Even Numbers
50 print("\n[TEST SET 1] Even Numbers")
51 assert classify_value(8) == "Even", "8 should be classified as Even"
52 print("✓ assert classify_value(8) == 'Even'")
53
54 assert classify_value(2) == "Even", "2 should be classified as Even"
55 print("✓ assert classify_value(2) == 'Even'")
56
57 assert classify_value(100) == "Even", "100 should be classified as Even"
58 print("✓ assert classify_value(100) == 'Even'")
59
60 assert classify_value(-4) == "Even", "-4 should be classified as Even"
61 print("✓ assert classify_value(-4) == 'Even'")
62
63 assert classify_value(-10) == "Even", "-10 should be classified as Even"
64 print("✓ assert classify_value(-10) == 'Even'")
65
66 assert classify_value(1000) == "Even", "1000 should be classified as Even"
67 print("✓ assert classify_value(1000) == 'Even'")
68
69
70 # TEST CASE 2: Odd Numbers
71 print("\n[TEST SET 2] Odd Numbers")
72 assert classify_value(7) == "Odd", "7 should be classified as Odd"
73 print("✓ assert classify_value(7) == 'Odd'")
74
75 assert classify_value(1) == "Odd", "1 should be classified as Odd"
76 print("✓ assert classify_value(1) == 'Odd'")
77
78 assert classify_value(99) == "Odd", "99 should be classified as Odd"
79 print("✓ assert classify_value(99) == 'Odd'")
80
81 assert classify_value(-3) == "Odd", "-3 should be classified as Odd"
82 print("✓ assert classify_value(-3) == 'Odd'")
83

```

```

84
85 assert classify_value(-15) == "Odd", "-15 should be classified as Odd"
86 print("✓ assert classify_value(-15) == 'Odd'")
87
88 assert classify_value(2023) == "Odd", "2023 should be classified as Odd"
89 print("✓ assert classify_value(2023) == 'Odd'")
90
91 # TEST CASE 3: Zero
92 print("\n[TEST SET 3] Zero")
93 assert classify_value(0) == "Zero", "0 should be classified as Zero"
94 print("✓ assert classify_value(0) == 'Zero'")
95
96
97 # TEST CASE 4: String Input (Invalid)
98 print("\n[TEST SET 4] String Input - Invalid")
99 assert classify_value("abc") == "Invalid Input", "'abc' should be Invalid Input"
100 print("✓ assert classify_value('abc') == 'Invalid Input'")
101
102 assert classify_value("123") == "Invalid Input", "'123' (string) should be Invalid Input"
103 print("✓ assert classify_value('123') == 'Invalid Input'")
104
105 assert classify_value("even") == "Invalid Input", "'even' should be Invalid Input"
106 print("✓ assert classify_value('even') == 'Invalid Input'")
107
108 assert classify_value("") == "Invalid Input", "empty string should be Invalid Input"
109 print("✓ assert classify_value('') == 'Invalid Input'")
110
111
112 # TEST CASE 5: Float Input (Invalid)
113 print("\n[TEST SET 5] Float Input - Invalid")
114 assert classify_value(3.14) == "Invalid Input", "3.14 (float) should be Invalid Input"
115 print("✓ assert classify_value(3.14) == 'Invalid Input'")
116
117 assert classify_value(2.0) == "Invalid Input", "2.0 (float) should be Invalid Input"
118 print("✓ assert classify_value(2.0) == 'Invalid Input'")
119
120 assert classify_value(-1.5) == "Invalid Input", "-1.5 (float) should be Invalid Input"
121 print("✓ assert classify_value(-1.5) == 'Invalid Input'")

```

```

149
150 assert classify_value(1000000) == "Even", "1000000 should be Even"
151 print("✓ assert classify_value(1000000) == 'Even'")
152
153 assert classify_value(-999999) == "Odd", "-999999 should be Odd"
154 print("✓ assert classify_value(-999999) == 'Odd'")
155
156 assert classify_value(-1000000) == "Even", "-1000000 should be Even"
157 print("✓ assert classify_value(-1000000) == 'Even'")
158
159
160 # TEST CASE 9: Edge Cases with Loop Logic Demonstration
161 print("\n[TEST SET 9] Edge Cases - Loop Logic Verification")
162 # Demonstrating conditional logic by testing boundary values
163 test_values = [
164     (1, "Odd"),
165     (2, "Even"),
166     (3, "Odd"),
167     (4, "Even"),
168     (5, "Odd"),
169     (-1, "Odd"),
170     (-2, "Even"),
171     (0, "Zero"),
172 ]
173
174 # Using loop to verify multiple edge cases
175 for value, expected in test_values:
176     result = classify_value(value)
177     assert result == expected, f"classify_value({value}) should return '{expected}', got '{res"
178     print(f"✓ assert classify_value({value}) == '{expected}'")
179
180
181 # =====
182 # DEMONSTRATION: Using loops for batch classification
183 # =====
184
185 print("\n" + "=" * 70)
186 print("BONUS: BATCH CLASSIFICATION USING LOOPS")
187 print("=" * 70)
188
189 test_numbers = list(range(-5, 11)) # Numbers from -5 to 10
190 print(f"\nClassifying range of numbers from -5 to 10:")

```

```

4
5 print("\n" + "=" * 70)
6 print("BONUS: BATCH CLASSIFICATION USING LOOPS")
7 print("=" * 70)
8
9 test_numbers = list(range(-5, 11)) # Numbers from -5 to 10
10 print("\nClassifying range of numbers from -5 to 10:")
11 print("-" * 70)
12
13 for num in test_numbers:
14     classification = classify_value(num)
15     print(f"classify_value({num:3d}) = {classification:15s}", end="")
16     if (num + 1) % 4 == 0:
17         print()
18     else:
19         print(" | ", end="")
20
21 print("\n")
22
23 # =====
24 # TEST SUMMARY
25 # =====
26
27 print("=" * 70)
28 print("ALL TESTS PASSED SUCCESSFULLY! ✓")
29 print("=" * 70)
30
31 print("\nClassification Rules Verified:")
32 print("  ✓ Even number detection (positive and negative)")
33 print("  ✓ Odd number detection (positive and negative)")
34 print("  ✓ Zero special case handling")
35 print("  ✓ String input rejection")
36 print("  ✓ Float input rejection")
37 print("  ✓ Boolean input rejection")
38 print("  ✓ None and complex type rejection")
39 print("  ✓ Conditional logic for classification")
40 print("  ✓ Loop-based batch processing capability")
41 print("\n")

```


OUTPUT:

```
=====
RUNNING EVEN-ODD & TYPE CLASSIFICATION TEST CASES
=====

[TEST SET 1] Even Numbers
✓ assert classify_value(8) == 'Even'
✓ assert classify_value(2) == 'Even'
✓ assert classify_value(100) == 'Even'
✓ assert classify_value(-4) == 'Even'
✓ assert classify_value(-10) == 'Even'
✓ assert classify_value(1000) == 'Even'

[TEST SET 2] Odd Numbers
✓ assert classify_value(7) == 'Odd'
✓ assert classify_value(1) == 'Odd'
✓ assert classify_value(99) == 'Odd'
✓ assert classify_value(-3) == 'Odd'
✓ assert classify_value(-15) == 'Odd'
✓ assert classify_value(2023) == 'Odd'

[TEST SET 3] Zero
✓ assert classify_value(0) == 'Zero'

[TEST SET 4] String Input - Invalid
✓ assert classify_value('abc') == 'Invalid Input'
✓ assert classify_value('123') == 'Invalid Input'
✓ assert classify_value('even') == 'Invalid Input'
✓ assert classify_value('') == 'Invalid Input'

[TEST SET 5] Float Input - Invalid
✓ assert classify_value(3.14) == 'Invalid Input'
✓ assert classify_value(2.0) == 'Invalid Input'
✓ assert classify_value(-1.5) == 'Invalid Input'

[TEST SET 6] Boolean Input - Invalid
✓ assert classify_value(True) == 'Invalid Input'
✓ assert classify_value(False) == 'Invalid Input'

[TEST SET 7] None, List, Dictionary - Invalid
✓ assert classify_value(None) == 'Invalid Input'

✓ assert classify_value(2) == 'Even'
✓ assert classify_value(3) == 'Odd'
✓ assert classify_value(4) == 'Even'
✓ assert classify_value(5) == 'Odd'
✓ assert classify_value(-1) == 'Odd'
✓ assert classify_value(-2) == 'Even'
✓ assert classify_value(0) == 'Zero'

=====
BONUS: BATCH CLASSIFICATION USING LOOPS
=====

Classifying range of numbers from -5 to 10:
-----
classify_value( -5) = Odd
classify_value( -4) = Even          | classify_value( -3) = Odd          | classify_value( -2) =
Even          | classify_value( -1) = Odd          |
classify_value(  0) = Zero          | classify_value(  1) = Odd          | classify_value(  2) =
Even          | classify_value(  3) = Odd          |
classify_value(  4) = Even          | classify_value(  5) = Odd          | classify_value(  6) =
Even          | classify_value(  7) = Odd          |
classify_value(  8) = Even          | classify_value(  9) = Odd          | classify_value( 10) =
Even          |

=====
ALL TESTS PASSED SUCCESSFULLY! ✓
=====

Classification Rules Verified:
✓ Even number detection (positive and negative)
✓ Odd number detection (positive and negative)
✓ Zero special case handling
✓ String input rejection
✓ Float input rejection
✓ Boolean input rejection
✓ None and complex type rejection
✓ Conditional logic for classification
✓ Loop-based batch processing capability
```

Task Description #3 (Palindrome Checker – Apply AI for String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a function `is_palindrome(text)` and implement the function.

- Requirements:

- o Ignore case, spaces, and punctuation.

- o Handle edge cases such as empty strings and single characters.

Example Assert Test Cases:

```
assert is_palindrome("Madam") == True
```

```
assert is_palindrome("A man a plan a canal Panama") ==  
True
```

```
assert is_palindrome("Python") == False
```

Expected Output #3:

- Function correctly identifying palindromes and passing all AI-generated tests

CODE:

```

tmp > palindrom_checker.py > ...
1  import re
2
3
4  def is_palindrome(text):
5      """
6      Check if a string is a palindrome, ignoring case, spaces, and punctuation.
7
8      Args:
9      |   text (str): The string to check
10
11      Returns:
12      |   bool: True if the string is a palindrome, False otherwise
13      """
14      # Remove non-alphanumeric characters and convert to lowercase
15      cleaned = re.sub(r'[^a-z0-9]', '', text.lower())
16
17      # Handle empty strings
18      if not cleaned:
19          return True
20
21      # Check if the cleaned string equals its reverse
22      return cleaned == cleaned[::-1]
23
24
25 # Test cases
26 if __name__ == "__main__":
27     # Basic palindrome examples
28     assert is_palindrome("Madam") == True
29     assert is_palindrome("A man a plan a canal Panama") == True
30     assert is_palindrome("Python") == False
31
32     # Edge cases - empty strings
33     assert is_palindrome("") == True
34     assert is_palindrome(" ") == True
35     assert is_palindrome("!!!") == True
36
37     # Edge cases - single characters
38     assert is_palindrome("a") == True
39     assert is_palindrome("A") == True
40     assert is_palindrome("1") == True

```

```

assert is_palindrome("!!!") == True

# Edge cases - single characters
assert is_palindrome("a") == True
assert is_palindrome("A") == True
assert is_palindrome("1") == True

# Additional test cases with various punctuation
assert is_palindrome("race a car") == False
assert is_palindrome("Was it a car or a cat I saw?") == True
assert is_palindrome("12321") == True
assert is_palindrome("12345") == False

# Case sensitivity tests
assert is_palindrome("RaceCar") == True
assert is_palindrome("LEVEL") == True

# Numbers and mixed characters
assert is_palindrome("A1b1A") == True
assert is_palindrome("1a2b2a1") == True

print("✓ All test cases passed!")

```

OUTPUT:

✓ All test cases passed!

Task Description #4 (BankAccount Class – Apply AI for Object-Oriented Test-Driven Development)

- Task: Ask AI to generate at least 3 assert-based test cases for a BankAccount class and then implement the class.
- Methods:
 - o deposit(amount)

o withdraw(amount)

o get_balance()

Example Assert Test Cases:

```
acc = BankAccount(1000)
```

```
acc.deposit(500)
```

```
assert acc.get_balance() == 1500
```

```
acc.withdraw(300)
```

```
assert acc.get_balance() == 1200
```

Expected Output #4:

- Fully functional class that passes all AI-generated assertions

CODE:

```
tmp > bank_account.py > % bankAccount > get_balance
1 class BankAccount:
2     """
3     A simple BankAccount class for managing deposits, withdrawals, and balance tracking.
4     """
5
6     def __init__(self, initial_balance=0):
7         """
8         Initialize a BankAccount with an optional initial balance.
9
10        Args:
11            initial_balance: The starting balance (default: 0)
12
13        Raises:
14            ValueError: If initial_balance is negative
15        """
16        if initial_balance < 0:
17            raise ValueError("Initial balance cannot be negative")
18        self.balance = initial_balance
19
20    def deposit(self, amount):
21        """
22        Deposit money into the account.
23
24        Args:
25            amount: The amount to deposit
26
27        Raises:
28            ValueError: If amount is negative or zero
29        """
30        if amount <= 0:
31            raise ValueError("Deposit amount must be positive")
32        self.balance += amount
33
34    def withdraw(self, amount):
35        """
36        Withdraw money from the account.
37
38        Args:
39            amount: The amount to withdraw
40
41        Raises:
```

```

        if amount > self._balance:
            raise ValueError("Insufficient funds for withdrawal")
        self._balance -= amount

    def get_balance(self):
        """
        Get the current account balance.

        Returns:
            The current balance
        """
        return self._balance

# AI-Generated Test Cases
print("Running BankAccount Class Tests...")
print("-" * 60)

# Test Case 1: Basic deposit and balance check
print("\nTest Case 1: Basic deposit and balance check")
acc1 = BankAccount(1000)
acc1.deposit(500)
assert acc1.get_balance() == 1500
print("✓ Test 1 passed: deposit(500) on balance 1000 = 1500")

# Test Case 2: Withdrawal and balance check
print("\nTest Case 2: Withdrawal and balance check")
acc1.withdraw(300)
assert acc1.get_balance() == 1200
print("✓ Test 2 passed: withdraw(300) on balance 1500 = 1200")

# Test Case 3: Multiple transactions
print("\nTest Case 3: Multiple transactions")
acc2 = BankAccount(2000)
acc2.deposit(1000)
acc2.withdraw(500)
acc2.deposit(250)
assert acc2.get_balance() == 2750
print("✓ Test 3 passed: Multiple transactions result in balance 2750")

```

```

78 print("\nTest Case 3: Multiple transactions")
79 acc2 = BankAccount(2000)
80 acc2.deposit(1000)
81 acc2.withdraw(500)
82 acc2.deposit(250)
83 assert acc2.get_balance() == 2750
84 print("✓ Test 3 passed: Multiple transactions result in balance 2750")
85
86 # Test Case 4: Account with zero initial balance
87 print("\nTest Case 4: Account with zero initial balance")
88 acc3 = BankAccount(0)
89 assert acc3.get_balance() == 0
90 acc3.deposit(100)
91 assert acc3.get_balance() == 100
92 print("✓ Test 4 passed: Zero-balance account initialized and deposit works")
93
94 # Test Case 5: Withdraw entire balance
95 print("\nTest Case 5: Withdraw entire balance")
96 acc4 = BankAccount(500)
97 acc4.withdraw(500)
98 assert acc4.get_balance() == 0
99 print("✓ Test 5 passed: Withdraw entire balance leaves 0")
100
101 # Test Case 6: Large amounts
102 print("\nTest Case 6: Large amounts handling")
103 acc5 = BankAccount(1000000)
104 acc5.deposit(500000)
105 assert acc5.get_balance() == 1500000
106 acc5.withdraw(250000)
107 assert acc5.get_balance() == 1250000
108 print("✓ Test 6 passed: Large amounts handled correctly")
109
110 # Test Case 7: Invalid negative initial balance
111 print("\nTest Case 7: Invalid negative initial balance")
112 try:
113     acc_invalid = BankAccount(-100)
114     print("X Test 7 failed: Should raise ValueError for negative balance")
115 except ValueError as e:
116     print(f"✓ Test 7 passed: Correctly raised ValueError: {e}")
117
118 # Test Case 8: Invalid negative deposit

```



```

112     try:
113         acc_invalid = BankAccount(-100)
114         print("X Test 7 failed: Should raise ValueError for negative balance")
115     except ValueError as e:
116         print(f"✓ Test 7 passed: Correctly raised ValueError: {e}")
117
118     # Test Case 8: Invalid negative deposit
119     print("\nTest Case 8: Invalid negative deposit")
120     acc6 = BankAccount(1000)
121     try:
122         acc6.deposit(-100)
123         print("X Test 8 failed: Should raise ValueError for negative deposit")
124     except ValueError as e:
125         print(f"✓ Test 8 passed: Correctly raised ValueError: {e}")
126
127     # Test Case 9: Invalid withdrawal exceeding balance
128     print("\nTest Case 9: Invalid withdrawal exceeding balance")
129     acc7 = BankAccount(100)
130     try:
131         acc7.withdraw(200)
132         print("X Test 9 failed: Should raise ValueError for insufficient funds")
133     except ValueError as e:
134         print(f"✓ Test 9 passed: Correctly raised ValueError: {e}")
135
136     # Test Case 10: Zero amount operations
137     print("\nTest Case 10: Zero amount operations")
138     acc8 = BankAccount(500)
139     try:
140         acc8.deposit(0)
141         print("X Test 10a failed: Should raise ValueError for zero deposit")
142     except ValueError as e:
143         print(f"✓ Test 10a passed: Correctly raised ValueError for zero deposit: {e}")
144
145     try:
146         acc8.withdraw(0)
147         print("X Test 10b failed: Should raise ValueError for zero withdrawal")
148     except ValueError as e:
149         print(f"✓ Test 10b passed: Correctly raised ValueError for zero withdrawal: {e}")
150
151     print("\n" + "-" * 60)
152     print("All tests completed successfully! ✓")

```

OUTPUT:

```

Running BankAccount Class Tests...
-----

Test Case 1: Basic deposit and balance check
✓ Test 1 passed: deposit(500) on balance 1000 = 1500

Test Case 2: Withdrawal and balance check
✓ Test 2 passed: withdraw(300) on balance 1500 = 1200

Test Case 3: Multiple transactions
✓ Test 3 passed: Multiple transactions result in balance 2750

Test Case 4: Account with zero initial balance
✓ Test 4 passed: Zero-balance account initialized and deposit works

Test Case 5: Withdraw entire balance
✓ Test 5 passed: Withdraw entire balance leaves 0

Test Case 6: Large amounts handling
✓ Test 6 passed: Large amounts handled correctly

Test Case 7: Invalid negative initial balance
✓ Test 7 passed: Correctly raised ValueError: Initial balance cannot be negative

Test Case 8: Invalid negative deposit
✓ Test 8 passed: Correctly raised ValueError: Deposit amount must be positive

Test Case 9: Invalid withdrawal exceeding balance
✓ Test 9 passed: Correctly raised ValueError: Insufficient funds for withdrawal

Test Case 10: Zero amount operations
✓ Test 10a passed: Correctly raised ValueError for zero deposit: Deposit amount must be positive
✓ Test 10b passed: Correctly raised ValueError for zero withdrawal: Withdrawal amount must be positive
-----
All tests completed successfully! ✓

```

Task Description #5 (Email ID Validation – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for a function `validate_email(email)` and implement the function.
- Requirements:

- o Must contain @ and .
- o Must not start or end with special characters.
- o Should handle invalid formats gracefully.

Example Assert Test Cases:

```
assert validate_email("user@example.com") == True
```

```
assert validate_email("userexample.com") == False
```

```
assert validate_email("@gmail.com") == False
```

Expected Output #5:

- Email validation function passing all AI-generated test cases and handling edge cases correctly.

CODE:

```
1 import re
2
3 def validate_email(email):
4     """
5     Validate an email address based on common requirements.
6
7     Requirements:
8     - Must contain @ symbol
9     - Must contain . (dot) symbol
10    - Must not start with special characters
11    - Must not end with special characters
12    - Must have text before @
13    - Must have text between @ and .
14    - Must have text after .
15
16    Args:
17        email: A string to validate as an email address
18
19    Returns:
20        True if email is valid, False otherwise
21    """
22    # Check if email is a string and not empty
23    if not isinstance(email, str) or not email:
24        return False
25
26    # Check if email contains @ and .
27    if '@' not in email or '.' not in email:
28        return False
29
30    # Check if email starts or ends with special characters
31    if email[0] in '@._-' or email[-1] in '@._-':
32        return False
33
34    # Split by @ to validate structure
35    if email.count('@') != 1:
36        return False
37
38    local_part, domain_part = email.split('@')
39
40    # Validate local part (before @)
41    if not local_part or len(local_part) > 64:
```

```

42 |     return False
43 |
44 |     # Local part should not start/end with special chars (except . is allowed inside)
45 |     if local_part[0] in '._-' or local_part[-1] in '._-':
46 |         return False
47 |
48 |     # Validate domain part (after @)
49 |     if not domain_part or len(domain_part) > 255:
50 |         return False
51 |
52 |     # Domain must contain exactly one . at proper position
53 |     if domain_part.count('.') < 1:
54 |         return False
55 |
56 |     # Split domain by .
57 |     domain_parts = domain_part.split('.')
58 |
59 |     # Check that domain parts are not empty
60 |     for part in domain_parts:
61 |         if not part or len(part) > 63:
62 |             return False
63 |
64 |     # Check that parts start/end with alphanumeric
65 |     for part in domain_parts:
66 |         if not part[0].isalnum() or not part[-1].isalnum():
67 |             return False
68 |
69 |     # Validate characters used in email
70 |     # Allowed: alphanumeric, dot, hyphen, underscore in local part
71 |     # Allowed: alphanumeric, dot, hyphen in domain part
72 |     valid_local_chars = set('abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789._-+')
73 |     valid_domain_chars = set('abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789.-')
74 |
75 |     if not all(c in valid_local_chars for c in local_part):
76 |         return False
77 |
78 |     if not all(c in valid_domain_chars for c in domain_part):
79 |         return False
80 |
81 |     return True

```

```

84 | # AI-generated Test Cases
85 | print("Running Email Validation Tests...")
86 | print("-" * 70)
87 |
88 | # Test Case 1: Valid email with standard format
89 | print("\nTest Case 1: Valid email with standard format")
90 | assert validate_email("user@example.com") == True
91 | print("✓ Test 1 passed: validate_email('user@example.com') == True")
92 |
93 | # Test Case 2: Invalid email - missing @
94 | print("\nTest Case 2: Invalid email - missing @")
95 | assert validate_email("userexample.com") == False
96 | print("✓ Test 2 passed: validate_email('userexample.com') == False")
97 |
98 | # Test Case 3: Invalid email - starts with @
99 | print("\nTest Case 3: Invalid email - starts with @")
100 | assert validate_email("@gmail.com") == False
101 | print("✓ Test 3 passed: validate_email('@gmail.com') == False")
102 |
103 | # Test Case 4: Valid email with numbers
104 | print("\nTest Case 4: Valid email with numbers")
105 | assert validate_email("user123@example456.com") == True
106 | print("✓ Test 4 passed: validate_email('user123@example456.com') == True")
107 |
108 | # Test Case 5: Valid email with underscore and dot in local part
109 | print("\nTest Case 5: Valid email with underscore and dot in local part")
110 | assert validate_email("john.doe_123@example.com") == True
111 | print("✓ Test 5 passed: validate_email('john.doe_123@example.com') == True")
112 |
113 | # Test Case 6: Invalid email - ends with dot
114 | print("\nTest Case 6: Invalid email - ends with dot")
115 | assert validate_email("user@example.") == False
116 | print("✓ Test 6 passed: validate_email('user@example.') == False")
117 |
118 | # Test Case 7: Invalid email - multiple @ symbols
119 | print("\nTest Case 7: Invalid email - multiple @ symbols")
120 | assert validate_email("user@exam@ple.com") == False
121 | print("✓ Test 7 passed: validate_email('user@exam@ple.com') == False")
122 |
123 | # Test Case 8: Invalid email - missing domain extension
124 | print("\nTest Case 8: Invalid email - missing domain extension")
125 | assert validate_email("user@example") == False

```

```

130 assert validate_email("") == False
131 print("✓ Test 9 passed: validate_email('') == False")
132
133 # Test Case 10: Invalid email - starts with dot
134 print("\nTest Case 10: Invalid email - starts with dot")
135 assert validate_email(".user@example.com") == False
136 print("✓ Test 10 passed: validate_email('.user@example.com') == False")
137
138 # Test Case 11: Invalid email - ends with special character
139 print("\nTest Case 11: Invalid email - ends with special character")
140 assert validate_email("user@example.com-") == False
141 print("✓ Test 11 passed: validate_email('user@example.com-') == False")
142
143 # Test Case 12: Valid email with hyphen in domain
144 print("\nTest Case 12: Valid email with hyphen in domain")
145 assert validate_email("user@ex-ample.com") == True
146 print("✓ Test 12 passed: validate_email('user@ex-ample.com') == True")
147
148 # Test Case 13: Invalid email - space in email
149 print("\nTest Case 13: Invalid email - space in email")
150 assert validate_email("user @example.com") == False
151 print("✓ Test 13 passed: validate_email('user @example.com') == False")
152
153 # Test Case 14: Invalid email - non-alphanumeric in domain start
154 print("\nTest Case 14: Invalid email - hyphen at domain start")
155 assert validate_email("user@-example.com") == False
156 print("✓ Test 14 passed: validate_email('user@-example.com') == False")
157
158 # Test Case 15: Valid email with plus sign in local part
159 print("\nTest Case 15: Valid email with plus sign in local part")
160 assert validate_email("user+tag@example.com") == True
161 print("✓ Test 15 passed: validate_email('user+tag@example.com') == True")
162
163 # Test Case 16: Invalid email - None type
164 print("\nTest Case 16: Invalid email - None type")
165 assert validate_email(None) == False
166 print("✓ Test 16 passed: validate_email(None) == False")
167
168 # Test Case 17: Invalid email - missing local part
169 print("\nTest Case 17: Invalid email - missing local part")
170 assert validate_email("@example.com") == False
171 print("✓ Test 17 passed: validate_email('@example.com') == False")

```

```

164 print("\nTest Case 16: Invalid email - None type")
165 assert validate_email(None) == False
166 print("✓ Test 16 passed: validate_email(None) == False")
167
168 # Test Case 17: Invalid email - missing local part
169 print("\nTest Case 17: Invalid email - missing local part")
170 assert validate_email("@example.com") == False
171 print("✓ Test 17 passed: validate_email('@example.com') == False")
172
173 # Test Case 18: Valid email with subdomain
174 print("\nTest Case 18: Valid email with subdomain")
175 assert validate_email("user@mail.example.co.uk") == True
176 print("✓ Test 18 passed: validate_email('user@mail.example.co.uk') == True")
177
178 # Test Case 19: Invalid email - consecutive dots
179 print("\nTest Case 19: Invalid email - consecutive dots in domain")
180 assert validate_email("user@example..com") == False
181 print("✓ Test 19 passed: validate_email('user@example..com') == False")
182
183 # Test Case 20: Valid simple email
184 print("\nTest Case 20: Valid simple email")
185 assert validate_email("a@b.c") == True
186 print("✓ Test 20 passed: validate_email('a@b.c') == True")
187
188 print("\n" + "-" * 70)
189 print("All 20 test cases completed successfully! ✓")
190

```


OUTPUT:

```
PS C:\tmp> python -c "exec(open('/tmp/email_validator.py').read())"
Running Email Validation Tests...
-----

Test Case 1: Valid email with standard format
â€œ Test 1 passed: validate_email('user@example.com') == True

Test Case 2: Invalid email - missing @
â€œ Test 2 passed: validate_email('userexample.com') == False

Test Case 3: Invalid email - starts with @
â€œ Test 3 passed: validate_email('@gmail.com') == False

Test Case 4: Valid email with numbers
â€œ Test 4 passed: validate_email('user123@example456.com') == True

Test Case 5: Valid email with underscore and dot in local part
â€œ Test 5 passed: validate_email('john.doe_123@example.com') == True

Test Case 6: Invalid email - ends with dot
â€œ Test 6 passed: validate_email('user@example.') == False

Test Case 7: Invalid email - multiple @ symbols
â€œ Test 7 passed: validate_email('user@example.com') == False

Test Case 8: Invalid email - missing domain extension
â€œ Test 8 passed: validate_email('user@example') == False

Test Case 9: Invalid email - empty string
â€œ Test 9 passed: validate_email('') == False

Test Case 10: Invalid email - starts with dot
â€œ Test 10 passed: validate_email('.user@example.com') == False

Test Case 11: Invalid email - ends with special character
â€œ Test 11 passed: validate_email('user@example.com-') == False

Test Case 12: Valid email with hyphen in domain
â€œ Test 12 passed: validate_email('user@ex-ample.com') == True

Test Case 13: Invalid email - space in email
â€œ Test 13 passed: validate_email('user @example.com') == False
```

```
PS C:\tmp> python -c "exec(open('/tmp/email_validator.py').read())"
â€œ Test 14 passed: validate_email('user@-example.com') == False

Test Case 15: Valid email with plus sign in local part
â€œ Test 15 passed: validate_email('user+tag@example.com') == True

Test Case 16: Invalid email - None type
â€œ Test 16 passed: validate_email(None) == False

Test Case 17: Invalid email - missing local part
â€œ Test 17 passed: validate_email('@example.com') == False

Test Case 18: Valid email with subdomain
â€œ Test 18 passed: validate_email('user@mail.example.co.uk') == True

Test Case 19: Invalid email - consecutive dots in domain
â€œ Test 19 passed: validate_email('user@example..com') == False

Test Case 20: Valid simple email
â€œ Test 20 passed: validate_email('a@b.c') == True

-----
All 20 test cases completed successfully! â€œ
PS C:\tmp>
```


